

CIBAO INTL, Dominican Republic

WMO: 784600

Lat: 19.406N

Lon: 70.605W

Elev: 172

StdP: 99.27

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.8	17.8	15.1	10.9	23.6	16.1	11.7	23.1	8.4	28.0	7.8	27.9	0.3	290	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	34.1	24.9	33.6	24.9	33.0	25.0	26.9	30.8	26.5	30.5	26.1	30.2	6.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.9	28.1	25.5	21.2	27.7	25.1	20.7	27.4	85.7	31.1	83.7	30.4	82.2	29.9	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	8.0	7.2	DB	14.2	35.4	1.5	1.1	13.1	36.2	12.2	36.9	11.4	37.6	10.3	38.4	(4)
(5)			WB	13.9	29.0	1.5	1.6	12.9	30.1	12.0	31.1	11.2	32.0	10.1	33.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	23.6	24.1	24.7	25.8	26.6	27.7	27.8	27.9	27.7	27.0	25.5	24.2	(6)	
	DBStd	2.00	1.29	1.39	1.78	1.36	1.15	1.00	0.98	1.13	0.97	1.15	1.34	1.35	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	5856	420	394	454	474	516	531	552	554	530	528	464	439	(10)	
	CDD18.3	2814	162	161	196	224	257	281	294	296	280	270	214	181	(11)	
	CDH23.3	22882	948	1053	1426	1709	2057	2621	2836	2828	2552	2270	1493	1089	(12)	
(13)	CDH26.7	8887	238	317	497	644	792	1133	1218	1228	1119	922	491	288	(13)	
(14)	Wind	WSAvg	3.2	2.6	3.0	3.1	3.4	3.6	4.1	4.1	3.8	3.0	2.7	2.2	2.5	(14)
(15) Precipitation	PrecAvg	1151	64	66	66	105	147	73	103	103	104	122	134	125	(15)	
	PrecMax	1570	135	177	174	228	388	175	250	248	203	232	373	328	(16)	
	PrecMin	837	7	6	14	6	23	23	24	38	15	14	16	19	(17)	
	PrecStd	217	34	51	41	63	99	42	58	53	56	60	83	77	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	31.9	33.1	33.2	33.8	34.2	34.2	35.2	34.9	34.1	32.2	31.2	(19)	
		MCWB	22.7	23.2	23.4	23.8	23.8	25.1	24.9	24.6	24.8	24.8	24.5	23.6	(20)	
	2%	DB	29.9	30.8	31.9	32.1	32.8	33.6	33.8	34.1	33.9	33.0	31.4	30.2	(21)	
		MCWB	22.4	22.6	22.8	23.4	24.4	25.0	25.0	25.0	25.2	25.1	24.4	23.5	(22)	
	5%	DB	29.0	29.9	30.9	31.2	31.9	33.0	33.0	33.1	33.0	32.1	30.8	29.2	(23)	
		MCWB	22.4	22.5	22.6	23.3	24.3	25.0	25.1	25.2	25.3	25.0	24.3	23.3	(24)	
	10%	DB	28.1	28.9	29.8	30.2	30.9	32.1	32.1	32.2	32.1	31.2	29.8	28.3	(25)	
		MCWB	22.5	22.2	22.5	23.2	24.3	25.0	25.0	25.2	25.2	25.0	24.3	23.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	24.7	25.2	26.0	26.6	27.2	27.2	27.6	27.2	26.9	26.1	25.5	(27)	
		MCDB	27.5	28.8	29.6	30.1	30.3	31.1	31.1	30.9	31.4	30.5	29.3	28.4	(28)	
	2%	WB	24.0	23.9	24.3	25.1	26.0	26.6	26.6	26.7	26.5	26.2	25.6	24.8	(29)	
		MCDB	26.8	27.7	28.4	29.3	29.5	30.6	30.5	30.9	30.7	30.0	28.8	27.7	(30)	
	5%	WB	23.5	23.4	23.7	24.5	25.5	26.1	26.1	26.2	26.1	25.9	25.1	24.3	(31)	
		MCDB	26.5	27.1	27.7	28.5	29.0	30.2	30.1	30.4	30.3	29.7	28.5	27.3	(32)	
	10%	WB	23.1	23.0	23.2	24.0	25.0	25.7	25.7	25.8	25.7	25.6	24.7	23.8	(33)	
		MCDB	26.1	26.7	27.1	27.9	28.6	29.7	29.7	29.9	29.9	29.3	28.0	26.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.9	10.1	9.9	9.2	9.6	9.5	9.5	10.0	9.3	8.9	9.2	(35)	
		MCDBR	10.7	11.1	11.4	10.7	10.2	10.4	10.3	10.4	10.9	10.2	10.1	10.1	(36)	
	5% WB	MCWBR	4.9	4.2	3.7	3.4	3.2	3.4	3.3	3.3	3.8	3.7	4.0	4.7	(37)	
		MCWBR	9.4	9.9	10.2	9.7	9.0	9.5	9.4	9.4	10.1	9.3	8.8	9.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.363	0.377	0.391	0.419	0.445	0.508	0.492	0.469	0.427	0.391	0.383	0.383	(40)	
		taud	2.544	2.493	2.450	2.400	2.348	2.119	2.171	2.264	2.412	2.521	2.532	2.501	(41)	
	Ebn at Noon		918	922	917	889	855	795	809	834	871	897	890	883	(42)	
		Edh at Noon	97	107	115	122	128	159	151	138	118	103	97	98	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.24	4.96	5.54	5.78	5.66	6.02	6.10	5.93	5.63	4.93	4.18	4.02	(44)	
	RadStd		0.29	0.28	0.39	0.43	0.37	0.33	0.31	0.24	0.35	0.36	0.36	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.27	N/A	N/A	N/A	N/A

Nomenclature: See separate page